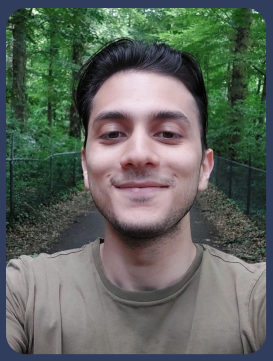

CEM KOÇ *Software Developer*

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Warsaw/Poland | LinkedIn | Portfolio | GitHub | YouTube

Artstation

EDUCATION

Futuregames - Warsaw,
Game Programming
09/2025 – present

Bahçeşehir University -Istanbul,
BCs in Computer Engineering
2019 – 2023

LANGUAGE

English: (Professional)

Turkish: (Native)

SKILLS

C#
Confident

C++
Confident

GLSL / Shader Programming
Can utilize when needed

Java
Can utilize when needed

TOOLS & FRAMEWORKS

Unity Engine
Confident

OpenGL
Intermediate

Unreal Engine
Learning

Arduino IDE
Mostly as a hobby

CAREER SUMMARY

Driven Software Developer with a BSc in Computer Science and a strong foundation in low-level programming and game engine architecture. Demonstrated history of engineering complex systems, spanning from developing a scholarship-winning robotic arm in high school to architecting a custom high-performance 2D game engine using C++ and OpenGL. Brings hands-on production experience from co-founding a studio incubated at BUG Lab TEKMER , alongside professional experience in AI and data science utilizing PyTorch. Currently refining advanced game development pipelines at Futuregames and contributing technically at Bosphorus Games.

WORK EXPERIENCE

Lead Programmer, *Bosphorus Games*
09/2025 – present | Istanbul - Turkiye

- Leading a programming team for a commercial PC title developed in Unity, overseeing both technical execution and collaborative workflows.
- Architecting and maintaining a highly scalable and readable codebase, establishing strict development standards that enable seamless feature integration by multiple programmers.
- Driving gameplay and systems programming by engineering robust underlying architectures, including the implementation of pushdown automata and hierarchical state machines for complex game logic.
- Serving as the project's technical artist, designing, writing, and optimizing custom shaders to meet specific visual requirements.

Keywords:
Team Lead / C# / Unity / Automata / Unity Shader Graph / HLSL

Co-Founder & Coder, *GoKo Studios*
02/2023 – 06/2024

- Co-founded a game development venture and led a 5-person team in the design and programming of a 2.5D fighting game using Unity and C#.
- Successfully pitched the studio to secure a 1-year incubation at BUG Lab TEKMER, Turkey's most renowned game development-focused incubation center.
- Authored core gameplay systems, actively upgrading coding standards to ensure a highly manageable, scalable, and readable codebase.
- Managed team operations and project timelines effectively, utilizing productivity tools such as Notion and Click-up.
- Demonstrated extreme adaptability by self-teaching 3D animation workflows in Blender to produce and integrate custom in-game animations when external resources were unavailable.

Keywords:
C# / Unity / Blender / 3D Animation / Notion & Click-up

Python
Can utilize when needed

MySQL
Can utilize when needed

INTERESTS

Video Games — (I enjoy playing a wide variety of game genres. If I were to name my favorites, they would be; action-adventure/RPG, tactical shooter, turn-based RPG, and sim/arcade racing.)

Motorsports/Vehicles — (Developing new technologies for vehicles to improve performance and safety.)

Music — (Developing Al solutions to match people's music taste with products on market.)

Film TV — (Using new technologies; AR, Game Engines etc. in film production.)

Other — Climbing/Bouldering, Playing Guitar, Tabletop Games

Data Scientist, *HCT Bilişim*
10/2022 – 06/2023

- Successfully balanced full-time university studies with a professional, hybrid work environment, requiring strong time management and self-directed problem-solving.
- Took ownership of the data pipeline, gathering requirements and adapting Al models specifically to meet the company's evolving desires for sentiment analysis.
- Researched multiple AI models and utilized PyTorch to develop custom solutions and modify existing architectures.
- Collected and parsed extensive data utilizing APIs from social media applications , applying complex data manipulation techniques with Pandas and NumPy.
- Managed database operations independently, ensuring seamless data flow by writing results to a MySQL server utilizing Python.

Keywords:
Python / MySql / Pandas / PyTorch

PROJECTS

Ovis Game Engine & Match Stranded,
C++ / Game Engine Development / OpenGL / GLSL / Premake
09/2024 – 02/2025

- Developed a custom 2D game engine from scratch utilizing C++ and OpenGL , requiring complex API design decisions to support large-scale, multi-platform development.
- Engineered high-performance rendering pipelines by implementing batch rendering, cache locality optimizations, and efficient data transfer methods.
- Successfully built and deployed a complete 2D game (Match Stranded) using the custom engine , demonstrating end-to-end framework viability and integrating Premake for build automation..

YouTube *GitHub*

Futuregames Game Projects
09/2025 – present

- Collaborated in cross-functional teams of up to 12 members—including designers, artists, and producers—to deliver highly polished games within strict 1-month production cycles.
- Secured 1st place for *Nana: Into the Banana* and 2nd place for *Polar Protocol* out of 8 competing games, as judged by industry professionals from prestige studios including CD Projekt Red and 11 Bit Studios.
- Acted as Programming Lead and Combat System Programmer for *Nana: Into the Banana*, designing scalable core mechanics.
- Engineered AI systems for *Polar Protocol*, utilizing and extending Unity's behavior tree packages for complex agent logic.

3D Scene Optimization with Octree, *C++ / OpenGL / GLSL / Octree*
08/2024 – 08/2024

- Engineered a C++ and OpenGL application designed to compute, visualize, and benchmark distance-check optimizations in densely populated 3D scenes.
- Implemented a custom Octree algorithm from scratch to significantly reduce algorithmic complexity and optimize memory management during spatial queries.
- Deepened expertise in the low-level rendering pipeline while writing highly optimized, hardware-efficient C++ code.

YouTube *GitHub*

Vampire Survivors Clone With Unity ECS, *Unity / C# / Unity ECS*
09/2025 – 10/2025

- Developed a high-density 3D top-down survival game, independently electing to architect the entire project using Unity's Entity Component System (ECS) to maximize performance and handle massive entity counts.
- Mastered the fundamentals of Data-Oriented Programming (DOP) by transitioning from traditional object-oriented paradigms to memory-efficient, data-driven pipelines.
- Engineered and optimized complex gameplay systems leveraging Unity's Data-Oriented Technology Stack (DOTS), ensuring high frame rates despite demanding 3D processing requirements.

YouTube *GitHub*